SpeakPay: Domain-Adaptive LoRA Fine-Tuning of Whisper for Low-Resource Nepali Financial Speech Recognition

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Abstract

Mobile payment applications in Nepal are graphically mediated and largely inaccessible to visually impaired users. This report presents SpeakPay, a voice-first digital wallet, and documents the central technical contribution: a controlled study of domain adaptation for low-resource financial speech recognition. We introduce NepFinSpeech-403, a 403-utterance dataset of Nepali financial voice commands (send, load, and balance operations spanning 237 unique numerals), and fine-tune Whisper large-v2 using Low-Rank Adaptation (LoRA) on this corpus. On a 60-utterance held-out test set, the domain-adapted model reduces Word Error Rate from 129.95% (zero-shot baseline) to 42.58% — a 67.2% relative reduction — and improves Devanagari numeral recognition accuracy from 0.0% to 56.8% averaged across utterances that contain numerals. The improvement is consistent at the individual-utterance level: the domain-adapted model outperforms the zero-shot baseline on 59 of 60 test utterances (sign test, $p = 1.7 \times 10^{-12}$). A per-intent breakdown shows the gain holds across all three command types (send, load, and a residual "other" category), though the "other" category reveals a labeling gap in our intent taxonomy — discussed in Section 6. The trained system, including a self-hosted inference API and a voice-first web application, is deployed and publicly accessible. All code, the dataset, model weights, and this report are released openly at `https://github.com/YOUR_USERNAME/speakpay`.

1. Introduction

Mobile wallets (eWallets) such as eSewa and Khalti have become the dominant mode of digital payment in Nepal. Their interfaces, however, are built entirely around visual interaction — icon-based navigation, on-screen forms, and graphical confirmation flows. For Nepal's visually impaired population (estimated at roughly 95,000 individuals [1]), this makes independent use of digital finance effectively impossible without sighted assistance.

Voice interaction is the natural accessibility solution, but two gaps stand in the way. First, existing Nepali Automatic Speech Recognition (ASR) systems are trained and evaluated on general speech [2, 3], not financial commands, which carry domain-specific demands: dense numeral sequences (transfer amounts, account balances), proper nouns (bank names, recipient names), and a narrow set of recurring sentence structures. Second, no public Nepali financial speech dataset existed prior to this work, making it impossible to even measure how badly general-purpose models perform on this domain, let alone improve on it.

This report documents the construction of such a dataset and a controlled comparison of zero-shot and domain-adapted ASR on it. The project began as an undergraduate group project with no rigorous evaluation, no public dataset, and no reproducible benchmark. This version is a complete independent rewrite: a newly assembled and characterized dataset, a reproducible LoRA fine-tuning pipeline, a statistically grounded evaluation, and a deployed application.

2. Related Work

Nepali ASR. Regmi and Bal [2] built an end-to-end Nepali ASR system with ESPnet on OpenSLR data, reporting a Character Error Rate (CER) of 10.3% on 159k general-domain utterances. Paudel et al. [3] used a CNN-Transformer architecture, achieving CER of 11.14% on a similarly general corpus. Both systems are evaluated on read speech or broad-domain utterances; neither reports performance on financial or transactional language, which differs substantially in vocabulary distribution and numeral density.

Whisper and parameter-efficient fine-tuning. Radford et al. [4] introduced Whisper, an encoder-decoder Transformer trained on 680,000 hours of weakly supervised multilingual audio. Whisper's zero-shot performance on Nepali is poor relative to high-resource languages; we quantify this directly for the financial domain in Section 5. Hu et al. [5] introduced LoRA, which freezes the base model and injects trainable low-rank decomposition matrices into attention and feed-forward layers, reducing trainable parameters by over 99% relative to full fine-tuning while matching or exceeding its quality on many tasks. This makes adaptation feasible on a single consumer GPU and, critically, on a dataset far too small to safely full-fine-tune a 1.55-billion-parameter model.

Accessible FinTech. Ilyasa et al. [6] built Visipay, an inclusive Indonesian eWallet evaluated via the System Usability Scale (SUS score 93). Zhou et al. [7] iteratively redesigned a cryptocurrency wallet for blind users, reporting reduced task completion time across design iterations. Neither system incorporates a custom or domain-adapted ASR component; both rely on existing screen-reader infrastructure layered over visual interfaces, rather than building a voice-first interaction model from the speech layer up.

Our work does not propose a new architecture or training method. It presents a controlled, statistically evaluated study of whether domain-specific LoRA adaptation closes the gap between a general-purpose ASR model and the specific requirements of financial voice interaction in a low-resource language — and ships the result as a working application.

3. Dataset: NepFinSpeech-403

3.1 Collection

Audio was collected through a purpose-built web platform. Contributors recorded spoken Nepali financial commands guided by written prompts, covering three operation types: fund transfers (recipient, amount, optionally a financial institution), wallet load/deposit commands, and balance enquiries. Transcripts were manually verified against the recorded audio.

3.2 Statistics

Table 1: NepFinSpeech-403 composition

Split	Samples	%
Train	283	70%
Validation	40	10%
Test (held-out)	60	15%
Total	**403**	100%

Intent (heuristic)	Count
Send	193 (47.9%)
Balance	61 (15.1%)
Load	56 (13.9%)
Other / unlabeled	93 (23.1%)
Unique Devanagari numerals	237

The "other" category is a heuristic residual class assigned by keyword matching, not a manually curated label; Section 6 shows it captures a genuine, previously unaccounted-for intent (third-person "funds received" statements), which we treat as a finding rather than noise.

4. Method

Base model. Whisper large-v2 [4], 1.55B parameters, encoder-decoder Transformer operating on log-Mel spectrograms.

Adaptation. LoRA [5] with rank $r = 32$, scaling $\alpha = 64$, applied to all query, key, value, output, and feed-forward projection matrices (`q_proj`, `k_proj`, `v_proj`, `out_proj`, `fc1`, `fc2`) across every encoder and decoder layer. This yields approximately 8M trainable parameters, under 0.6% of the base model, while the remaining 99.4% stays frozen.

Why LoRA, not full fine-tuning. With only 283 training utterances, full fine-tuning of a 1.55B-parameter model risks both overfitting and catastrophic forgetting of the base model's general speech competence. LoRA's parameter efficiency is not merely a compute convenience here — it is the difference between a feasible and an infeasible experiment at this dataset scale.

Training configuration. fp16 precision, no quantization, single consumer GPU (RTX 3060, 12GB), effective batch size 16 via gradient accumulation, learning rate 1×10^{-4}. Full configuration in `training/scripts/config.py` of the accompanying repository.

5. Results

5.1 Aggregate benchmark

All models are evaluated on the same 60-utterance held-out test split, never seen during training (fixed seed for reproducibility).

Table 2: Benchmark results on the NepFinSpeech-403 test set

Model	WER% ↓	CER% ↓	NumAcc% ↑
Whisper large-v2 (zero-shot)	129.95	92.32	0.0
Whisper large-v2 (general Nepali FT)	—	—	—
Whisper large-v2 + LoRA (ours)	**42.58**	**16.95**	**56.8**

The general-domain Nepali fine-tune baseline failed to load during this benchmark run; see Section 7 (Limitations) for discussion. The remaining comparison is between the zero-shot base model and our domain-adapted model.

The zero-shot WER exceeds 100%, which indicates the model's output is, on average, longer than the reference — consistent with insertion-heavy hallucination rather than simple mistranscription. Manual inspection of zero-shot predictions confirms this: outputs frequently contain phonetically plausible but semantically incoherent Devanagari, including malformed numeral sequences and occasionally repeated phrase fragments. Domain adaptation does not merely improve transcription accuracy; it recovers basic output coherence in a domain where the base model fails qualitatively, not just quantitatively.

5.2 Per-utterance significance

Aggregate WER can be dominated by a small number of catastrophic failures. We therefore additionally report a paired, per-utterance comparison: for each of the 60 test utterances, we compute WER under both the zero-shot and domain-adapted models and compare directly.

Table 3: Paired per-utterance comparison (domain-adapted vs. zero-shot)

Utterances where domain-adapted model improves	59 / 60
Utterances where zero-shot is better	0 / 60
Tied	1 / 60
Mean per-utterance WER reduction	87.7 percentage points
Sign test (two-sided, $n = 59$ non-tied pairs)	$p = 1.7 \times 10^{-12}$

The improvement is not driven by a handful of outliers: the domain-adapted model is better on essentially every individual utterance in the test set, and the sign test rejects the null hypothesis of no systematic difference at an extremely high confidence level. Given the small absolute sample size ($n = 60$), we report the non-parametric sign test rather than relying on aggregate means alone, which can be more sensitive to distributional assumptions and outlier influence.

5.3 Per-intent breakdown

Table 4: WER by command intent (heuristic classification)

Intent	N	Zero-shot WER%	Ours WER%
Send	30	143.6	45.8
Load	8	124.1	34.8
Other	22	114.5	41.1

The improvement holds consistently across all three groups, with the largest absolute gain on the *send* category — both the most frequent intent in the dataset and the one with the highest numeral and proper-noun density, consistent with the hypothesis that domain adaptation disproportionately helps exactly the structurally hardest utterances.

6. Error Analysis

Qualitative inspection of the "other" category (Table 4) surfaces a genuine taxonomy gap rather than annotation noise. A recurring pattern — third-person statements describing funds already received, e.g. अर्चना श्रेष्ठले कुमारीबैंकमारु ४७५० प्राप्त गरेकाछन् ("Archana Shrestha received Rs. 4750 at Kumari Bank") — does not map cleanly onto the send/load/balance taxonomy used elsewhere in this work, since it is neither a command nor strictly a balance query. We did not anticipate this pattern during dataset design; it appears to have entered the corpus because some contributors interpreted "describe a financial transaction" prompts more broadly than intended.

We treat this as a useful negative result: an intent classifier trained naively on the present three-class scheme would silently misclassify this "receive" pattern, and any production deployment of the application's NLP layer should add a fourth intent class or explicitly reject out-of-taxonomy utterances rather than forcing a best-effort match. We did not retrain the application's intent parser to add this class in the current version; this is identified as future work in Section 7.

A second, smaller pattern in the "other" category is unrelated demographic statements (e.g. stating one's age) appearing in place of an expected financial command. The frequency is too low ($n = 4$ of 60 test utterances) to support a strong claim about cause, but it is consistent with occasional prompt-following drift during data collection rather than a model artifact, since it appears identically in both zero-shot and domain-adapted transcriptions of the same audio.

7. System Architecture and Deployment

The application is a Next.js web app deployed on Vercel, communicating with a Supabase-backed PostgreSQL database (authentication, wallet balances, and an atomic SQL transfer function to prevent race conditions on concurrent transactions). Voice commands are captured client-side, sent to a self-hosted FastAPI inference server (Hugging Face Spaces, Docker runtime) running the LoRA-adapted model, and the resulting transcript is parsed by a two-stage rule-and-confidence intent classifier before a spoken confirmation is presented to the user. Self-hosting the inference endpoint, rather than depending solely on a third-party serverless inference API, was adopted after observing unreliable cold-start behavior under the free tier of hosted inference — a practical deployment lesson as relevant as the modeling result itself for a system intended for real accessibility use.

8. Limitations and Future Directions

Incomplete baseline comparison. The general-domain Nepali fine-tune baseline failed to load during the benchmark run reported here (Table 2). The present comparison is therefore zero-shot vs. domain-adapted only; a complete three-way comparison against a strong general-domain fine-tune is necessary before claiming the gain is specific to financial-domain adaptation rather than to fine-tuning on *any* Nepali data. This is the highest-priority follow-up experiment.

Dataset scale. 403 utterances (283 for training) is small relative to standard ASR fine-tuning corpora. The results here should be read as a feasibility demonstration — domain adaptation provides large relative gains even at this scale — not as a claim of state-of-the-art absolute performance. NumAcc of 56.8% indicates substantial remaining error on numeral transcription, the single most safety-critical metric for a payment application; this is not yet production-ready accuracy.

Single-annotator transcription. Transcripts were manually verified but not independently double-annotated; no inter-annotator agreement statistic is available.

Intent taxonomy gap. As discussed in Section 6, a recognizable "funds received" utterance pattern is not represented in the current three-class intent scheme. Future work should add this as an explicit fourth class and re-evaluate the application's NLP intent parser specifically (as distinct from the ASR transcription quality evaluated in this report).

No formal accessibility evaluation. No usability study has yet been conducted with visually impaired users, the system's intended audience. A System Usability Scale (SUS) study, following the methodology used by Ilyasa et al. [6], is planned future work and would substantially strengthen the application-level (as opposed to model-level) contribution of this project.

Training dynamics not logged. Per-checkpoint validation WER/loss during training was not preserved for this report; a re-run with logged checkpoints would allow direct assessment of whether the model overfit within the chosen training step budget.

9. Conclusion

We presented NepFinSpeech-403, the first domain-specific Nepali financial speech dataset, and showed that LoRA fine-tuning of Whisper large-v2 on this small corpus produces a large, statistically robust improvement over the zero-shot baseline: a 67.2% relative WER reduction, recovery of basic output coherence in a domain where the base model otherwise produces incoherent transcriptions, and improvement on 59 of 60 individual test utterances ($p = 1.7 \times 10^{-12}$). The gain is consistent across command types, and error analysis surfaced a genuine gap in our intent taxonomy rather than merely confirming existing assumptions. The system is deployed as a working voice-first eWallet application, demonstrating that the distance between a research result and a usable accessibility tool is bridgeable with modest additional engineering. The principal remaining gap — a complete comparison against a general-domain fine-tuned baseline — is identified as the immediate next experiment.

The full codebase, training pipeline, dataset, model weights, and this report are available at `https://github.com/YOUR_USERNAME/speakpay`.

References

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